

AIMS AND OBJECTIVES

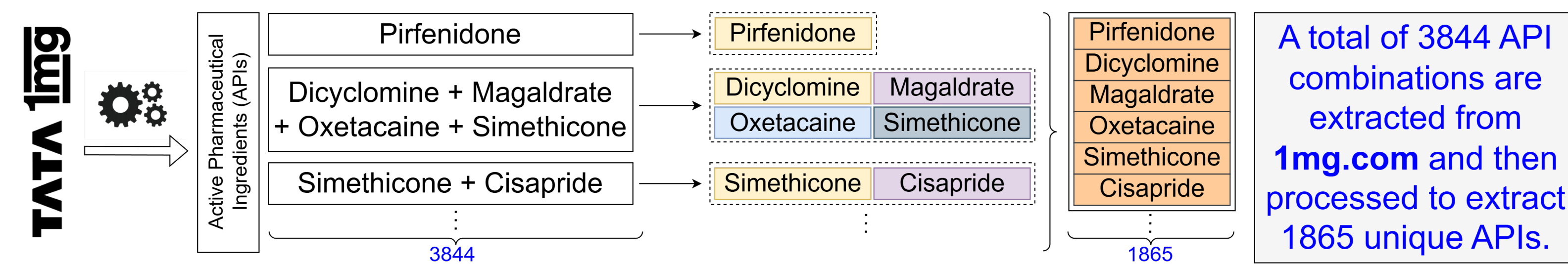
This project aims to analyse antibiotic prescription patterns in India for outpatient febrile respiratory illnesses by leveraging classical Natural Language Processing (NLP) techniques and open-source Large Language Models (osLLMs) to extract, standardise and structure data from unstructured clinical notes to the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM).

- To develop a comprehensive and standardised drug knowledge base of medication terms relevant to the Indian healthcare context.
- To design and implement NLP, osLLM-based algorithms & data pipelines to extract, normalise and map clinical entities from unstructured clinical notes to OMOP CDM.

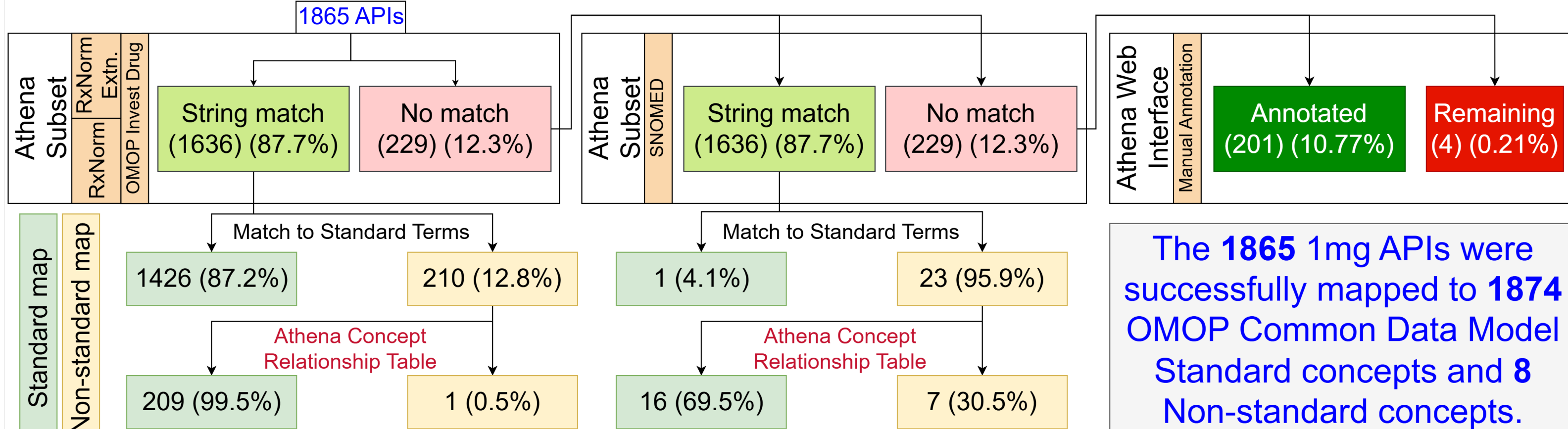
METHODS AND RESULTS

- Active Pharmaceutical Ingredients (APIs) from **1mg.com** were extracted and mapped to OMOP Standard Concepts using OHDSI standardised vocabularies. Extracted API OMOP concepts were linked to NRCeS CDCI (June 2025) and FDA GSRS UNII (July 2025) datasets via RxCUI and other identifiers.
- Antibiotics were classified by mapping the WHO AWaRe APIs to OMOP concepts and assigning AWaRe categories to extracted APIs and brand drug names.
- Clinical terms were extracted from Basic & SOAP notes using NLP and osLLMs; analysed pairwise term & algorithm overlaps, and compared prompt-engineering vs. parameter-efficient fine-tuned osLLMs for drug-name extraction from clinical prescription notes.

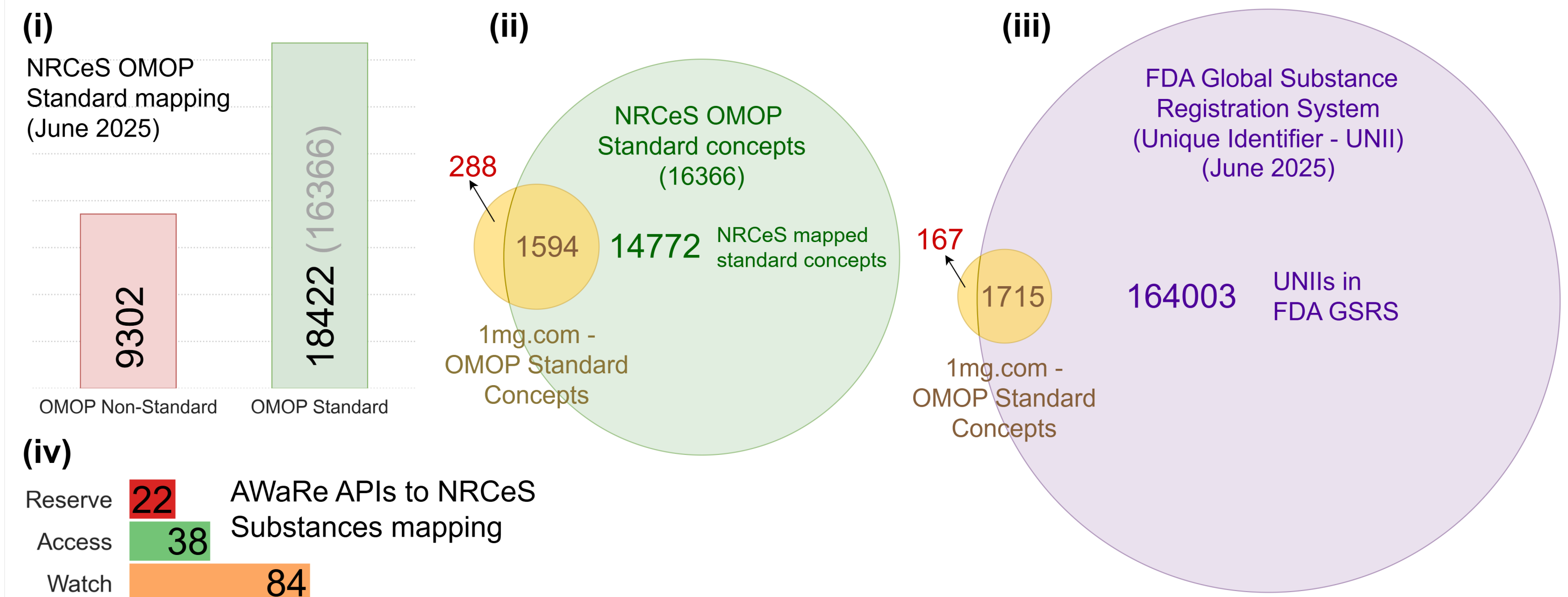
1A. Extracting Active Pharmaceutical Ingredients from 1mg.com



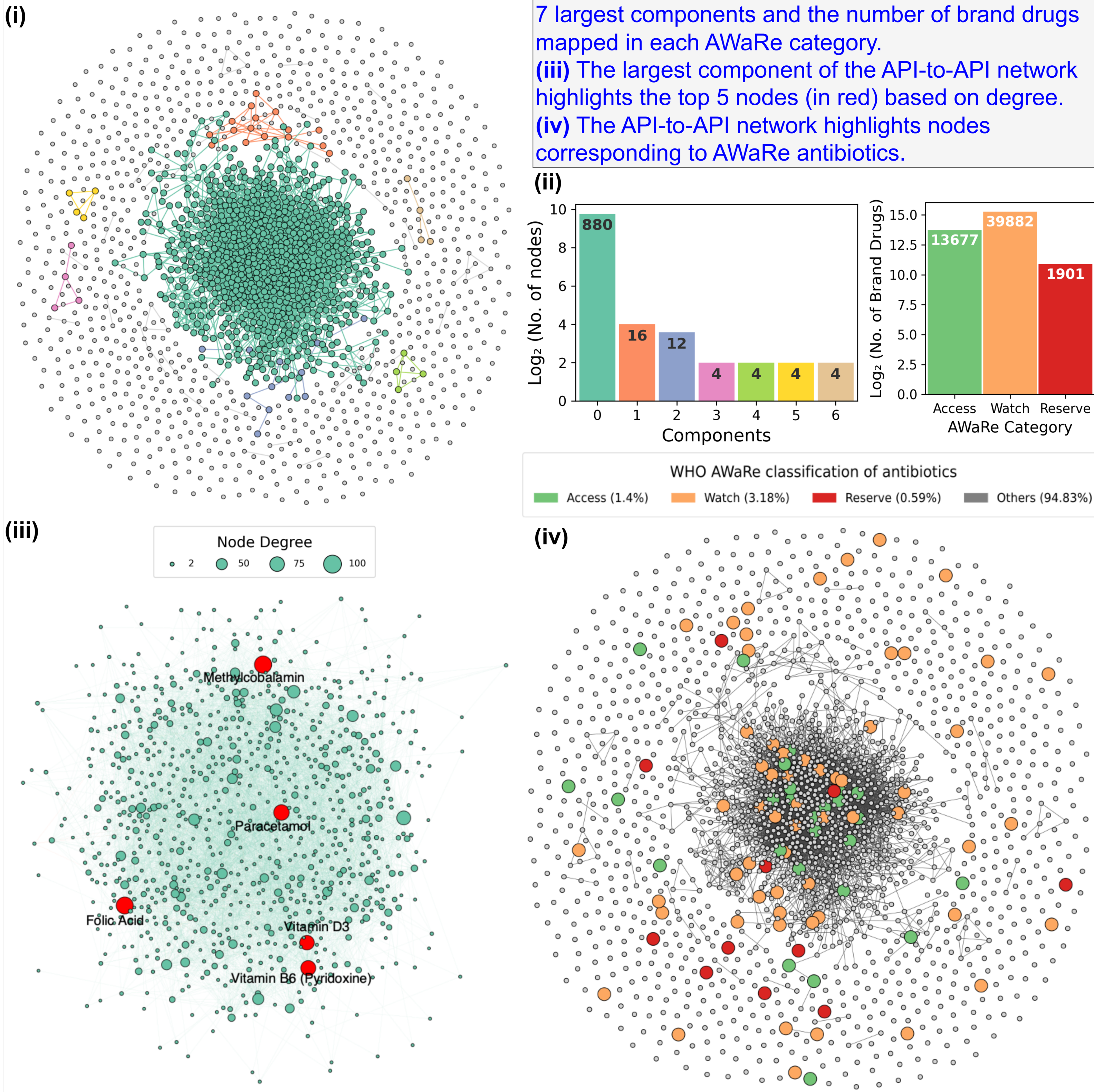
1B. Methodology for mapping 1mg.com APIs to OMOP Common Data Model concepts



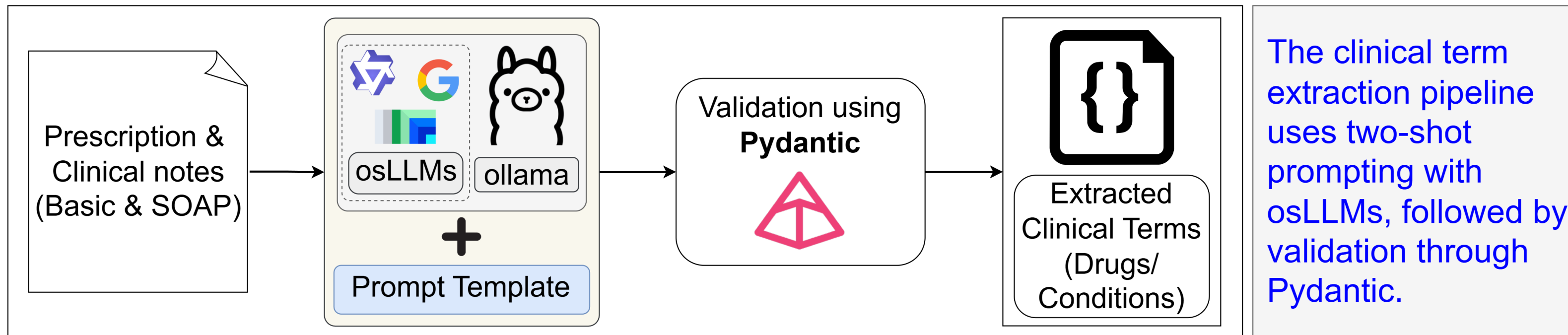
1C. Mapping 1mg.com OMOP concepts with NRCeS CDCI & FDA GSRS



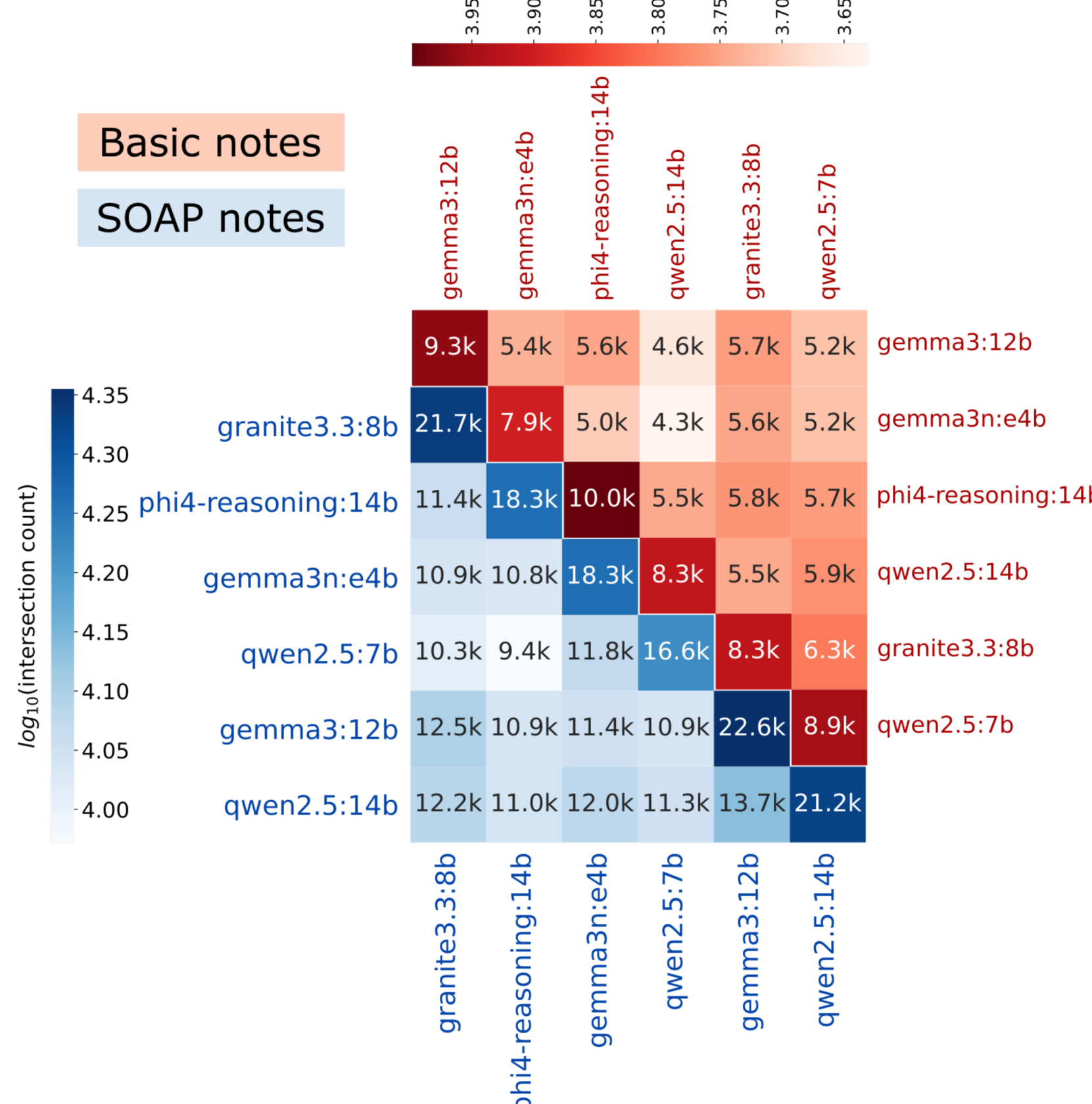
2. Visualisation of API-to-API network & WHO AWaRe antibiotic prevalence in Indian drug knowledge base



3A. Extraction of clinical terms using open-source Large Language Models (osLLMs)



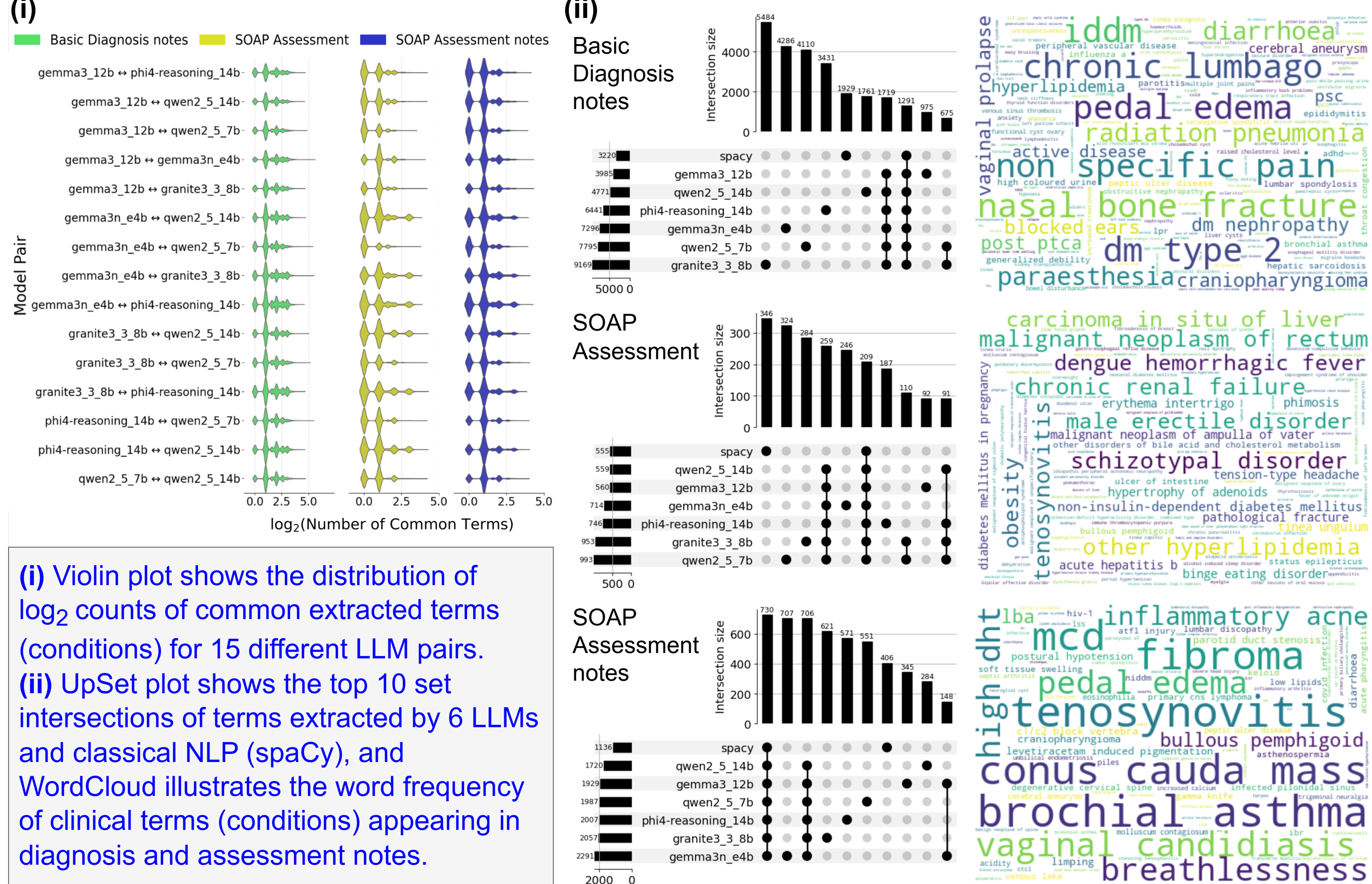
3B. Pairwise LLM set intersection heatmap of extracted drug terms



3C. WordCloud of extracted drug terms for Basic & SOAP prescription notes



3D. Extraction of clinical terms (conditions) using osLLM & spaCy for diagnosis & assessment notes



3E. Performance of LLMs for drug name extraction task on basic prescription notes

Model	F1 Score	Hallucinated	Error Rate
Mistral v0.3 7B	0.81 ± 0.29	0.88 ± 1.27	0.15 ± 0.29
Gemma 2 9B	0.84 ± 0.31	0.38 ± 0.76	0.16 ± 0.31
Llama 3.1 8B	0.81 ± 0.33	0.49 ± 0.95	0.19 ± 0.33
Qwen 2 7B	0.77 ± 0.32	0.89 ± 1.21	0.20 ± 0.32
DeepSeek-R1 7B	0.76 ± 0.35	0.75 ± 1.14	0.24 ± 0.35

Model	F1 Score	Hallucinated	Error Rate
Mistral v0.3 7B	0.82 ± 0.35	0.28 ± 0.57	0.18 ± 0.36
Gemma 2 9B	0.90 ± 0.23	0.21 ± 0.44	0.10 ± 0.24
Llama 3.1 8B	0.95 ± 0.18	0.13 ± 0.38	0.06 ± 0.20
Meditron 7B	0.93 ± 0.19	0.21 ± 0.63	0.06 ± 0.19
Llama 3 8B	0.92 ± 0.22	0.18 ± 0.52	0.08 ± 0.22
Mistral v0.1 7B	0.92 ± 0.22	0.27 ± 0.89	0.08 ± 0.22
Medalpaca 7B	0.80 ± 0.36	0.72 ± 1.70	0.21 ± 0.36
Llama 2 7B	0.65 ± 0.46	0.45 ± 0.56	0.35 ± 0.46

FUTURE DIRECTION

- To utilise the API knowledge base specific to medications available in India, and standardised & harmonised clinical data with the knowledge base will help in understanding antibiotic prescription patterns in febrile respiratory illnesses and other diseases.
- To expand the schema to link APIs with other major global databases such as PubChem, DrugBank, ChEBI and ChEMBL, apart from openFDA, to enable broader interoperability and cross-database analytics.

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